

LING324 Cognitive Linguistics

Professor Christopher Hart



Module Info

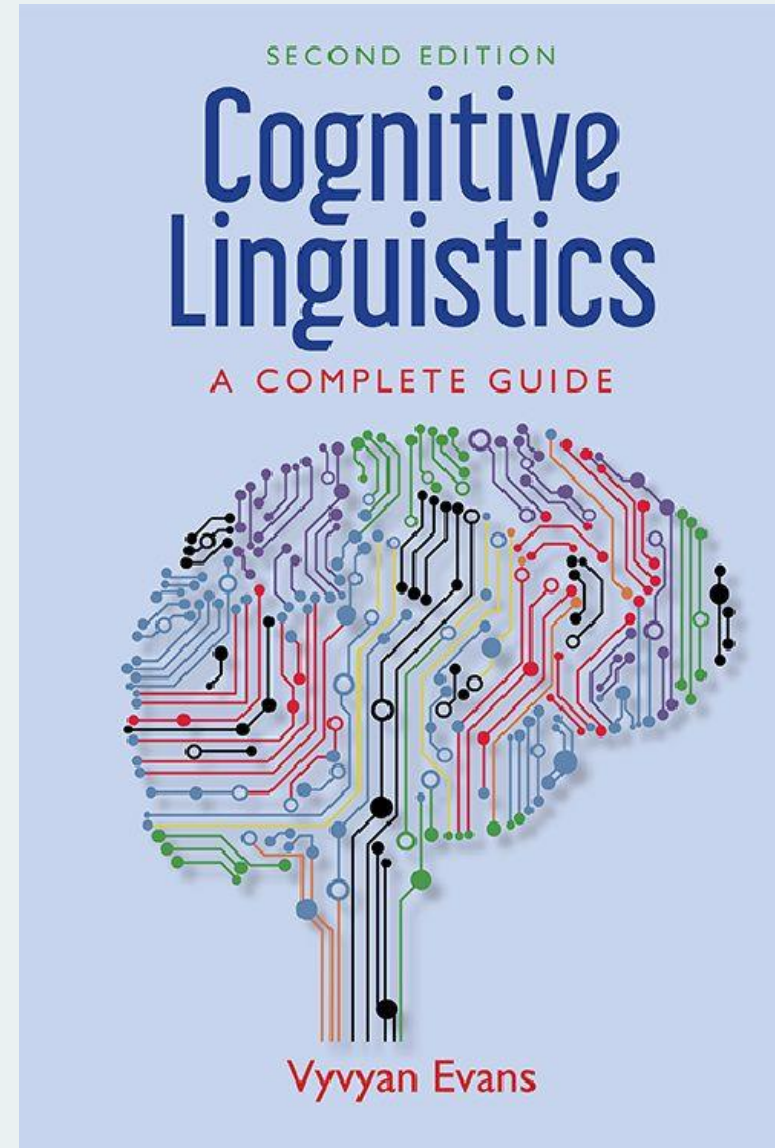
- Convenor: Professor Christopher Hart
- Email: c.hart@lancaster.ac.uk
- Student hour: tbc (C78)
- Seminars: discussion questions, exercises
- Assessment:
 - 2 x group presentations (20% each)
 - 1 x 2500-3000 word essay (60%)

Module Overview

- Introduction
- Radial categories and prototypes
- Frames
- Image schemas
- Conceptual metaphors
- READING WEEK
- Scenes and events
- Cognitive grammar
- Constructions
- Applications

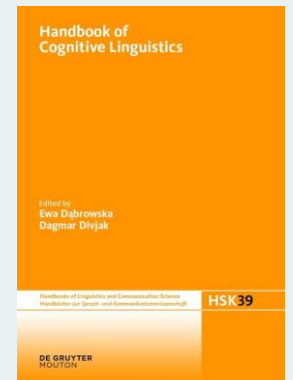
Reading

- Chapters from Evans (2019)
- See Module Handbook (Moodle)



What is Cognitive Linguistics?

[A]n approach to language study based on the assumptions that our linguistic abilities are firmly rooted in our general cognitive abilities, that meaning is essentially conceptualisation, and that grammar is shaped by usage. (Dąbrowska & Divjak 2015: 1)



- Developed originally in the 1980s in opposition to dominant approaches of the time:
 - Formal and truth-conditional semantics (e.g. Montague 1974)
 - Generative transformational grammar (e.g. Chomsky 1957, 1965)

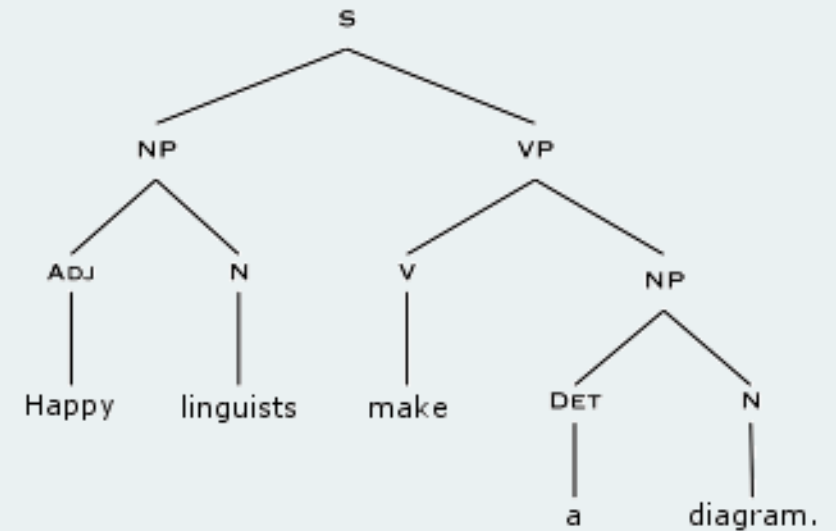
What is Cognitive Linguistics?

- Formal and truth-conditional semantics
 - Meaning reduced to real-world conditions that must obtain in order that a sentence be true
 - Meaning in natural language can be captured and represented using formal (mathematical) metalanguages such as **Propositional** or **Predicate Logic**
 - Inspired by formalism of generative grammar with roots in analytic philosophy

$\wedge$	<i>and</i> [conjunction]
$\vee$	<i>or</i> [disjunction]
$\Rightarrow$	<i>implies</i> [implication]
$\neg$	<i>not</i> [negation]
$\forall$	<i>For all</i>
$\exists$	<i>There exists</i>

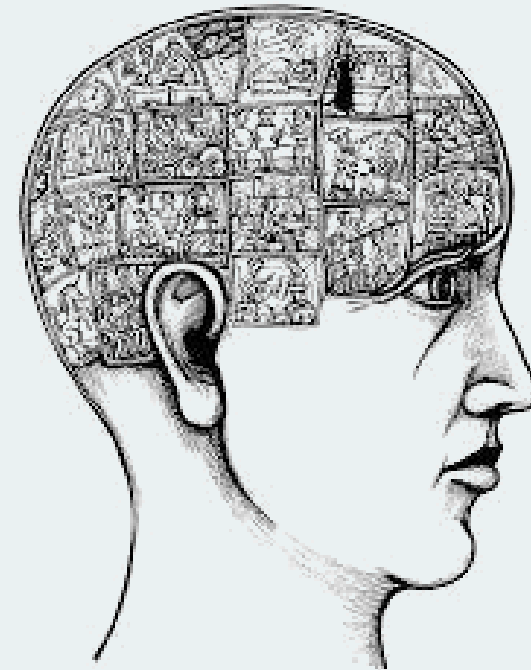
What is Cognitive Linguistics?

- Generative transformational grammar:
 - Language an autonomous cognitive faculty or **module** cut off from other cognitive abilities
 - Modularity of mind hypothesis (Fodor 1983)
 - Grammar (syntax) main area of investigation, considered as independent of meaning
 - Inventory of (universal and innate) **rules** accounting for realisation of **deep structure** in **surface structures**
 - Response to behaviourism



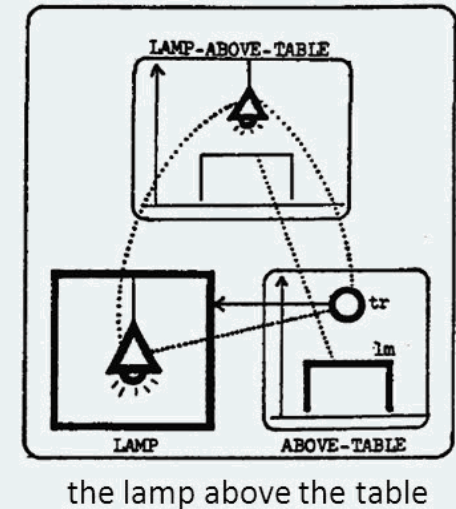
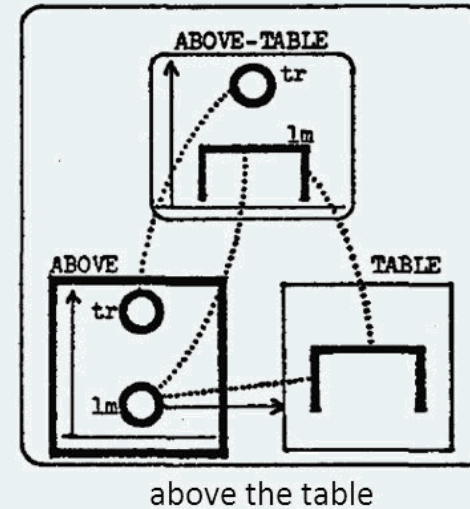
What is Cognitive Linguistics?

- Modularity of mind hypothesis:
 - Mind made up of distinct systems (module) uniquely dedicated to specific tasks
 - Colour perception, perception of shape and spatial relations, language
 - Domain-specificity, obligatory firing, rapid speed, information encapsulation



What is Cognitive Linguistics?

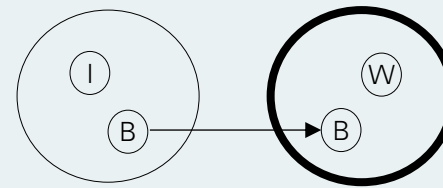
- Instead, Cognitive Linguistics:
 - Rejects the idea of an autonomous language module
 - Presents a unified account of grammar, semantics and pragmatics
 - Considers all elements of language to be meaningful, including grammatical constructions
 - Offers a naturally intuitive and psychologically plausible model of linguistic description
 - Considers linguistic structure to emerge through usage
 - Is functionalist and relativist in orientation



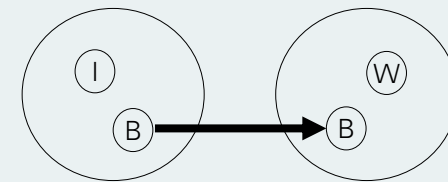
What is Cognitive Linguistics?

- In illustration of the semantic properties of grammatical constructions, consider the double-object or ditransitive construction vs. the prepositional indirect-object construction (the **dative alternation**):

1. (a) Ivy gave Wilbur a book
(b) Ivy gave a book to Wilbur
2. (a) Ivy gave the fence a coat of paint
(b) ?Ivy gave a coat of paint to the fence
3. (a) ?Wilbur sent America a parcel
(b) Wilbur sent a parcel to America



ditransitive



prepositional

(Langacker 1986: 14)

- Ditransitive construction focusses on the **result** of the process while prepositional construction foregrounds **movement** within it (Langacker 1990: 13)
- 2(b) odd because coat of paint not normally conceived as moveable object while 3(a) odd because America does not come into possession of the parcel as a result of the process
- No deep structure from which synonymous forms derived nor one derived from the other but alternate constructions which exist because meaningfully distinct

What is Cognitive Linguistics?

COGNITIVE SCIENCE 10, 1–40 (1986)

An Introduction to Cognitive Grammar

RONALD W. LANGACKER

University of California, San Diego

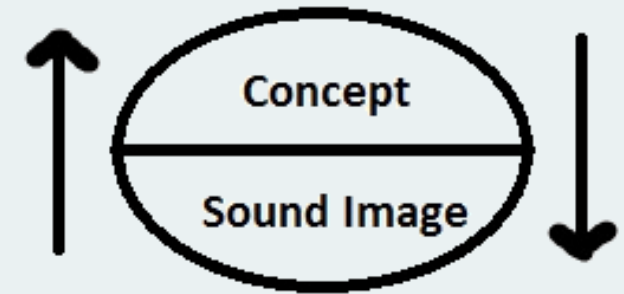
Cognitive grammar takes a nonstandard view of linguistic semantics and grammatical structure. Meaning is equated with conceptualization. Semantic structures are characterized relative to cognitive domains, and derive their value by construing the content of these domains in a specific fashion. Grammar is not a distinct level of linguistic representation, but reduces instead to the structuring and symbolization of conceptual content. All grammatical units are symbolic: Basic categories (e.g., noun and verb) are held to be notionally definable, and grammatical rules are analyzed as symbolic units that are both complex and schematic. These concepts permit a revealing account of grammatical composition with notable descriptive advantages.

Despite the diversity of contemporary linguistic theory, certain fundamental views enjoy a rough consensus and are widely accepted without serious question. Points of general agreement include the following: (a) language is a self-contained system amenable to algorithmic characterization, with sufficient autonomy to be studied in essential isolation from broader cognitive concerns; (b) grammar (syntax in particular) is an independent aspect of linguistic structure distinct from both lexicon and semantics; and (c) if meaning falls within the purview of linguistic analysis, it is properly described by some type of formal logic based on truth conditions. Individual theorists would doubtlessly qualify their assent in various ways, but (a)–(c) certainly come much closer than their denials to representing majority opinion.

The standard transformational analysis of these sentences treats them as synonymous and derives them from a common deep structure: depending on the particular choice of deep structure, *to* is either deleted or inserted transformationally, and the nonsubject nominals are permuted in the course of deriving the surface form of either (a) or (b). Cognitive grammar does not posit abstract deep structures, and neither sentence type is derived from the other – they are claimed instead to represent alternate construals of the profiled event. [1](a) and (b) differ in meaning because they employ subtly different images to structure the same conceived situation. (Langacker 1986: 14)

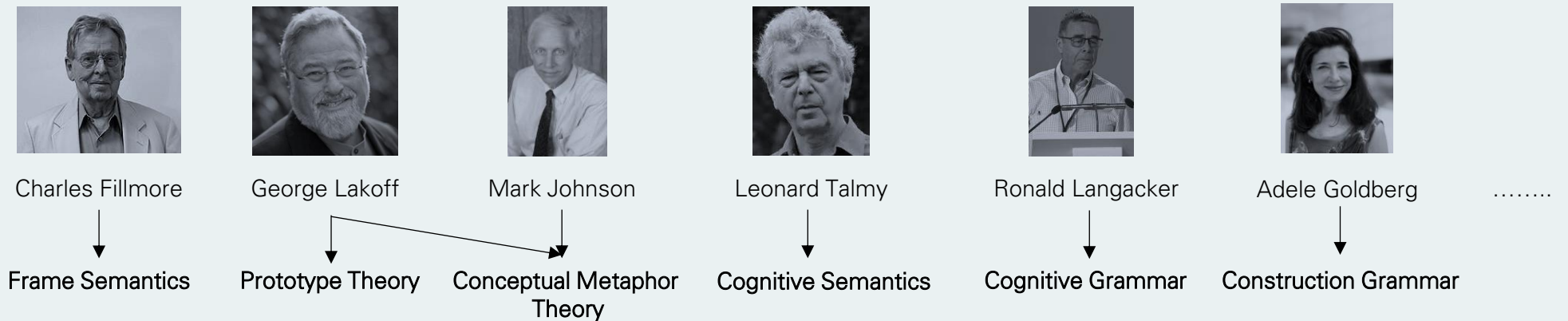
What is Cognitive Linguistics?

- Returns to the Saussurian notion of the **linguistic sign** which relates linguistic forms to concepts (meanings)
- Radical claim of CL is that this extends to units both smaller and larger than lexical units (words), including phonemes, morphemes and grammatical constructions
- CL also revitalises the **linguistic relativity hypothesis** as conventional ways of encoding specific situations differ across languages (see Pederson 2012)



Key Names and Frameworks

- CL made up of multiple frameworks associated with various key figures, e.g.:



- Frameworks coalesce around common set of assumptions concerning nature of language and way it should be described/investigated

Key Claims

- Claim 1: Language is NOT an autonomous cognitive module
 - Language relies on domain-general cognitive processes
 - The processes that support language are not unique to language but operate also in other areas of cognitive experience such as memory, perception, imagination and reason
 - Structures and processes of language not propositional or algorithmic but **conceptual** and imagistic
 - Analogues in other areas of cognitive experience and can be described appropriately in same terms
 - **Cognitive commitment:** “a commitment to make one’s account of human language accord with what is generally known about the mind and the brain, from other disciplines as well as our own” (Lakoff 1990: 40)

Key Claims

Construal operations as instances of general cognitive processes:

Attention/salience	
Selection	Profiling, metonymy
Scope	Scope of predication, search domains, accessibility
Scalar adjustment	Generality, specificity
Dynamicity	Fictive motion, summary/sequential scanning
Judgement/comparison	
Categorisation	
Metaphor	
Figure/ground	
Perspective/situatedness	
Viewpoint	Vantage point, orientation
Deixis	Spatiotemporal, epistemic, empathic
Subjectivity/objectivity	
Constitution/Gestalt	
Structural schematisation	Individuation, topological/geometric, scale
Force-dynamics	
Relationality	

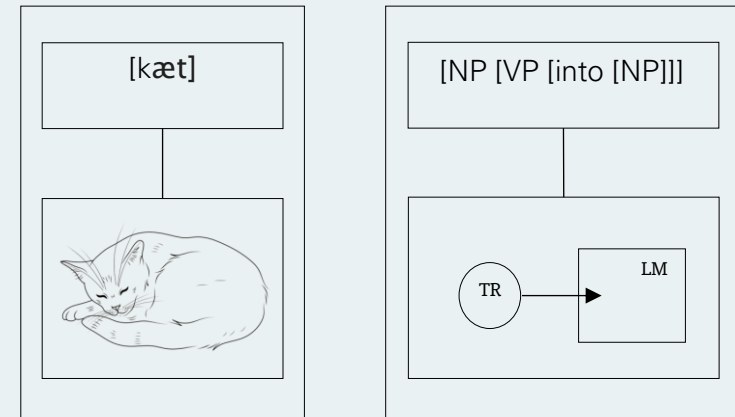
(adapted from Croft & Cruse 2004: 46)

Key Claims

- Claim 2: The language system itself is NOT modular
 - The same processes operate all levels of language – from phonemes up
 - Categorisation, metaphor, image-schematic patterning, perspective and attention discernible at all levels of language
 - Continuity of lexicon, grammar and pragmatics
 - **Generalisation Commitment:** “a commitment to characterising the general principles governing all aspects of human language” (Lakoff 1990: 40)
 - The Symbolic Thesis ➡

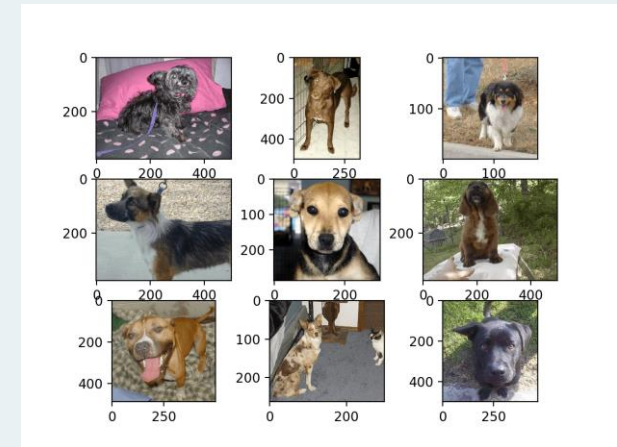
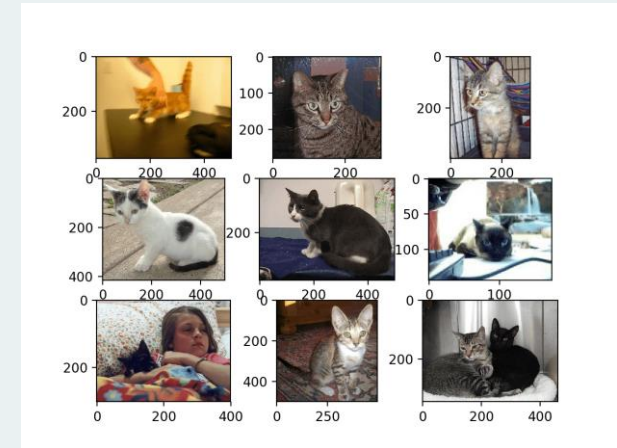
Key Claims

- **The Symbolic Thesis** (Langacker 2008)
 - Continuity of grammar and lexicon – Lexicon, morphology and syntax “form a continuum that is fully reducible to assemblies of symbolic structures” (Langacker 2008: 15)
 - Language comprises system of **symbolic assemblies** or **form-meaning pairings**
 - Difference between lexical and grammatical units resides in degrees of schematicity and complexity



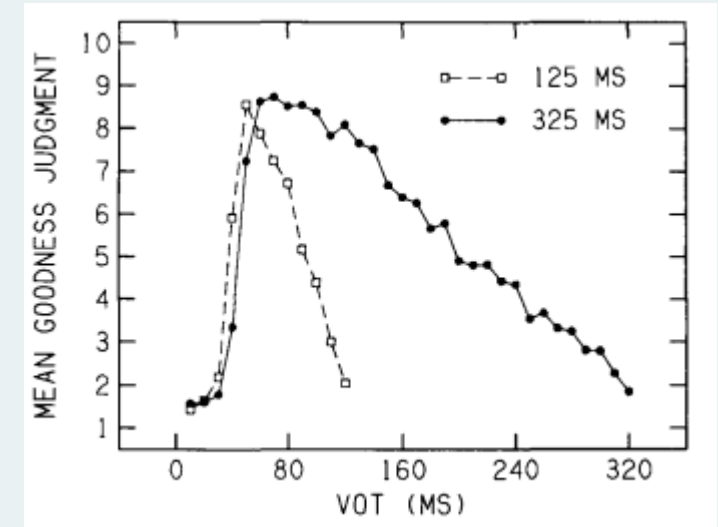
Key Claims

- E.g. Categorisation
 - General cognitive strategy used to organise experience into finite set of classes
 - Categories not absolute or strictly criterial but **fuzzy** and subject to **prototype** effects (Rosch 1975, 1977)
 - Applies to linguistic categories too, at all levels



Key Claims

- E.g. Categorisation
 - Phonological categories like VOICED (+/-) described as binary but some sounds are 'more voiced' or 'more voiceless' than others (Jaeger and Ohla 1984)
 - E.g nasal consonants /m/ and /n/ judged as more voiced than voiced fricatives /v, ð, z/ and voiced stops /b, d, g/
 - Phonemes realised differently in different speech contexts (e.g. depending on VOT and syllable duration)
 - Certain instances judged as better examples than others (Miller and Volaitis 1989)



Mean category-goodness ratings for /p/ by VOT and syllable duration (Miller and Volaitis 1989: 510)

Key Claims

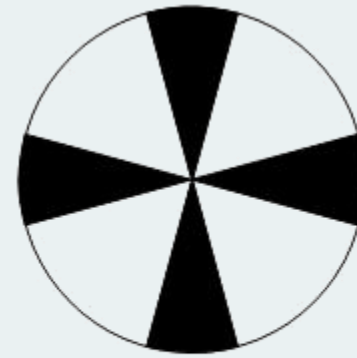
- E.g. Categorisation
 - Lexical categories like NOUN and VERB similarly gradient – some nouns and verbs more noun- and verb-like in their behaviours than others (Taylor 2003; Aarts 2007)
 - E.g. transitive verbs typically permit nominalisation but not all do:
 1. (a) David reads books (b) David is a readerer of books
 2. (a) David understands maths (b) ?David is an understanderer of maths
 3. (a) David eats meat (b) ?David is an eaterer of meat
 - E.g. most nouns accept pluralisation but not all (e.g. count nouns) while some nouns only occur in plural form
 4. (a) bookss (b) *musics (c) whereabouts

Key Claims

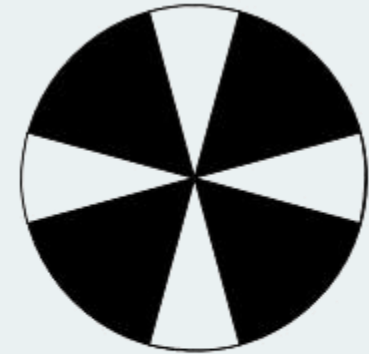
- E.g. Categorisation
 - Grammatical constructions also subject to prototype effects
 - E.g. prototypical examples of the double-object (ditransitive) and indirect-object prepositional constructions:
 1. Ivy gave Wilbur the book > Ivy gave the book to Wilbur
 2. Ivy gave it to Wilbur > Ivy gave Wilbur it
 - Prototypical ditransitive takes full NP for THEME while pronominal theme more typical of prepositional dative construction

Key Claims

- E.g. Attention
 - General cognitive ability that allows us to distinguish foreground from background
 - Observed in visual and auditory perception
 - Manifested in language too
(Langacker 2008; Talmy 2000)



A

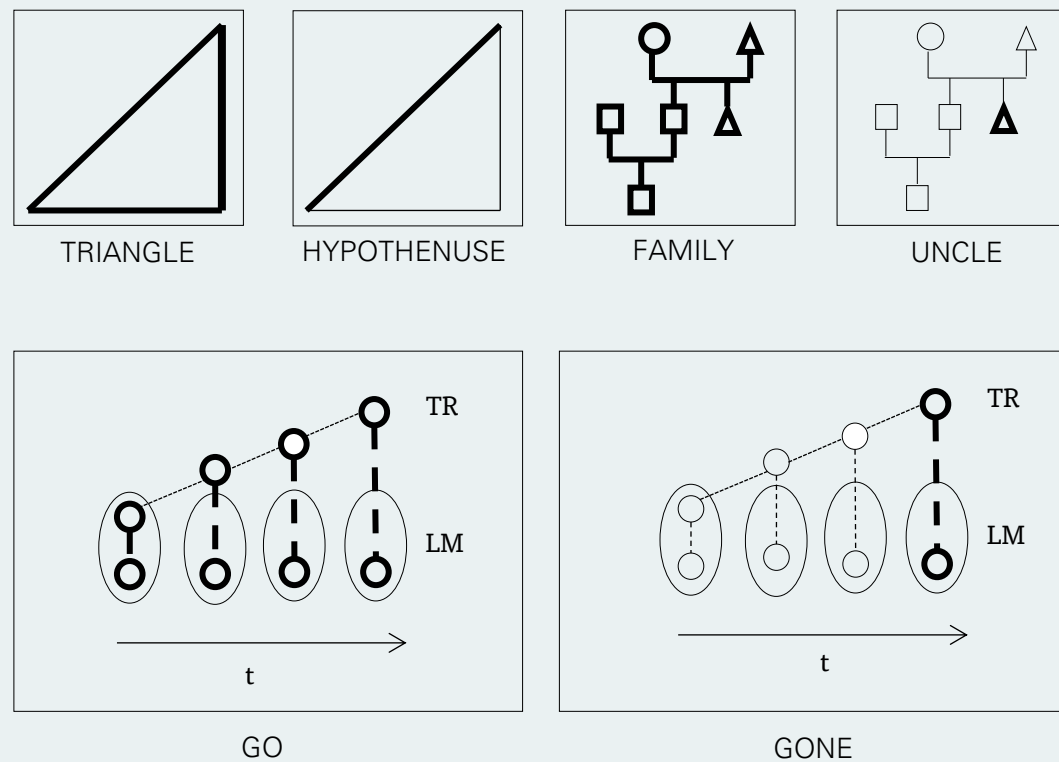


B

Key Claims

- E.g. Attention

- Lexical units derive meaning by highlighting for specific attention some **profile** within a **base** structure
- Grammatical elements likewise characterizable in terms of profile-base relations
- “The semantic contribution of the past participle inflection is to restrict the profile of the stem ...to its final state” (Langacker 1986: 7)



(adapted from Langacker 1986: 7)

Key Claims

- Claim 3: Language is polysemous
 - Polysemy not restricted to word meanings but fundamental a feature of language exhibited at all levels

run	'move fast' 'manage' 'provide' 'use' 'flow'
make	'prepare' 'force' 'appoint' 'reach' 'represent'

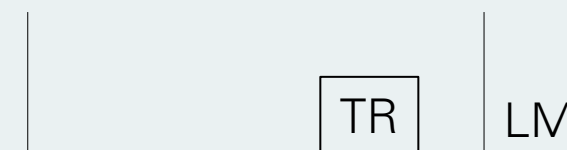
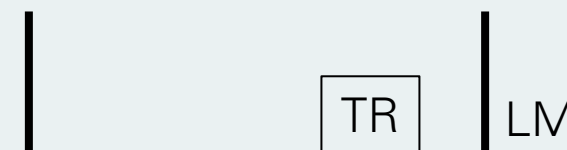
Ability (animate)	teacher cleaner
Ability (inanimate)	toaster blender
Attribute (origin)	villager southerner
Attribute (quality)	teatotaler

A. 'X CAUSES Y TO RECEIVE Z' (central sense) Example: Joe gave Sally the ball.
B. 'Conditions of satisfaction imply 'X CAUSES Y TO RECEIVE Z' Example: Joe promised Bob a car.
C. 'X CAUSES Y NOT TO RECEIVE Z' Example: Joe refused Bob a cookie.
D. 'X ACTS TO CAUSE Y TO RECEIVE Z at some future point in time' Example: Joe bequeathed Bob a fortune.
E. 'X ENABLES Y TO RECEIVE Z' Example: Joe permitted Chris an apple.
F. 'X INTENDS TO CAUSE Y TO RECEIVE Z' Example: Joe baked Bob a cake.

Polysemy of ditransitive construction (after Goldberg 1995: 75)

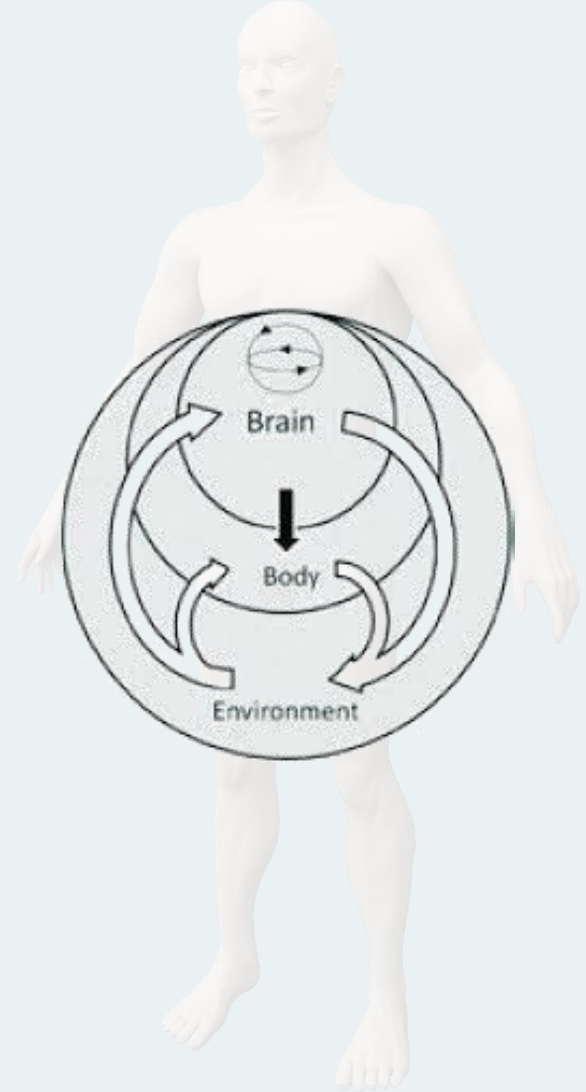
Key Claims

- **Claim 4: Language encodes experience**
 - There is no such thing as synonymy
 - Words may share the same denotation but differ in their connotation
 - **Principle of no synonymy:** If two constructions are syntactically distinct, they must be semantically or pragmatically distinct (Goldberg 1995: 67)
 - E.g. ditransitive versus prepositional structure
 - Alternate linguistic formulations impose contrasting **construals** on (evoke contrasting images of) the situations they describe
 - (a) The car parked in the street
 - (b) The car parked on the street



Key Claims

- **Claim 5: Language is grounded in experience (i)**
 - The meanings attached to linguistic expressions are rooted in our experience of the world, including experiences we have of our bodies
 - **Image schemas**
 - **Embodiment thesis:** features of the mind, including language, are shaped by the kind of bodies we have and the experiences they afford of the physical environment (e.g. Gibbs 2005)
 - **Experiential realism:** an objective world exists but our experience of it, to which language relates, is subjective, mediated by our bodies (Lakoff 1987)



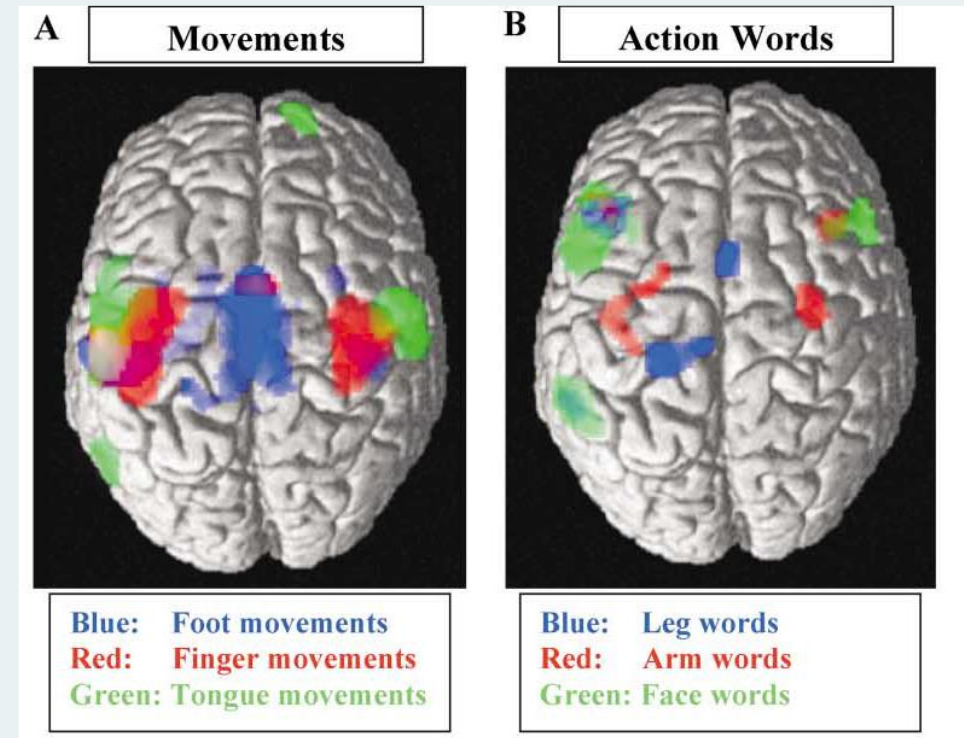
Key Claims

- Claim 5: Language is grounded in experience (ii)
 - The meanings attached to linguistic expressions are rooted in our experience of the world, including experiences we have as members of a given culture
 - **Frames**
 - **Encyclopaedic thesis:** The meaning of a linguistic expression extends beyond its immediate denotation to access vast repositories of structured world knowledge with diminishing degrees of salience
 - Linguistic knowledge *is* world knowledge
 - No principled distinction between semantics and pragmatics



Key Claims

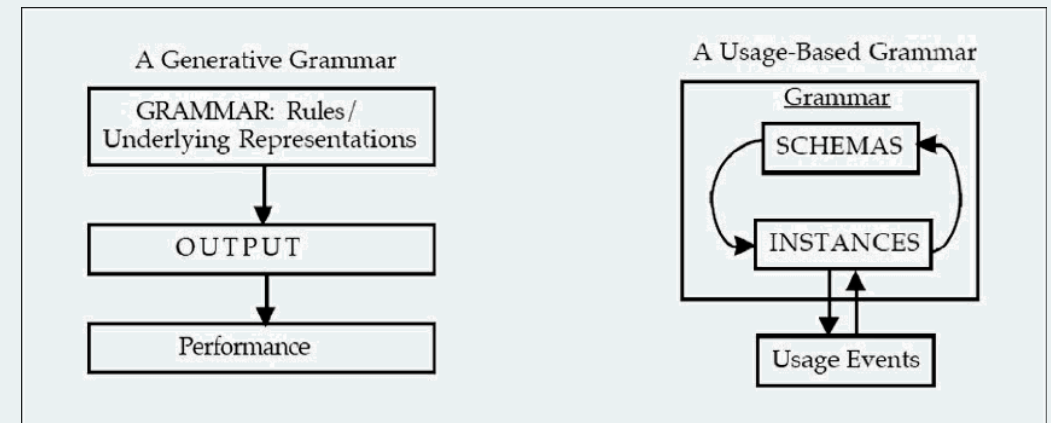
- Claim 5: Language is grounded in experience (iii)
 - Meaning is not just schematic but involves **simulations** (Bergen 2012)
 - Mental recreations of perceptual and motor experiences in understanding language
 - FMRI studies show areas of the brain activated in perceiving an object or performing an action also activated in response to sentences describing those objects and actions



(Hauk et al. 2004)

Key Claims

- Claim 6: Linguistic structure emerges through usage
 - **Usage-based thesis:** Linguistic knowledge derived from usage-events by process of **abstraction**
 - Recurrent form-meaning pairings become **conventionalised** and **entrenched** in system of symbolic assemblies
 - Form follows function!



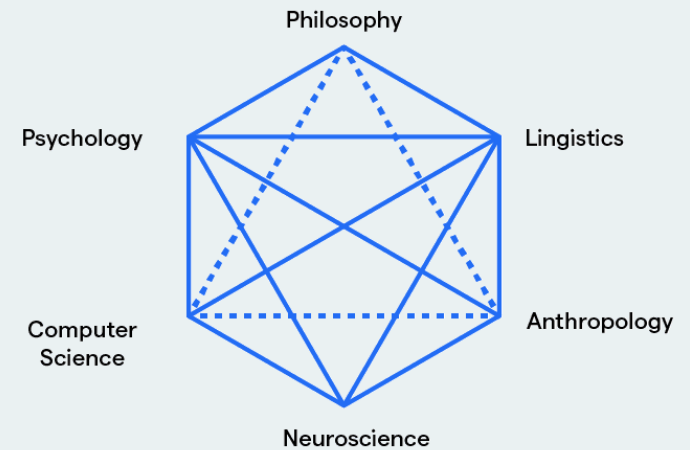
(Kemmer & Israel 1994: 168)

Connections with other Disciplines

- In order to fulfil its cognitive commitment, CL makes connections with other disciplines in cognitive science, including:

- Psychology
- Anthropology
- Neuroscience

Cognitive Science



Empirical Developments

- **Introspection** - “conscious attention directed by a language user to particular aspects of language manifest in her own cognition” (Talmy 2007: xii)
- Increasing use of empirical methods (Gonzalez-Marquez et al. 2007):
 - Corpus linguistics
 - Psycholinguistics
 - Neuroscience
 - Multimodality inc. gesture studies



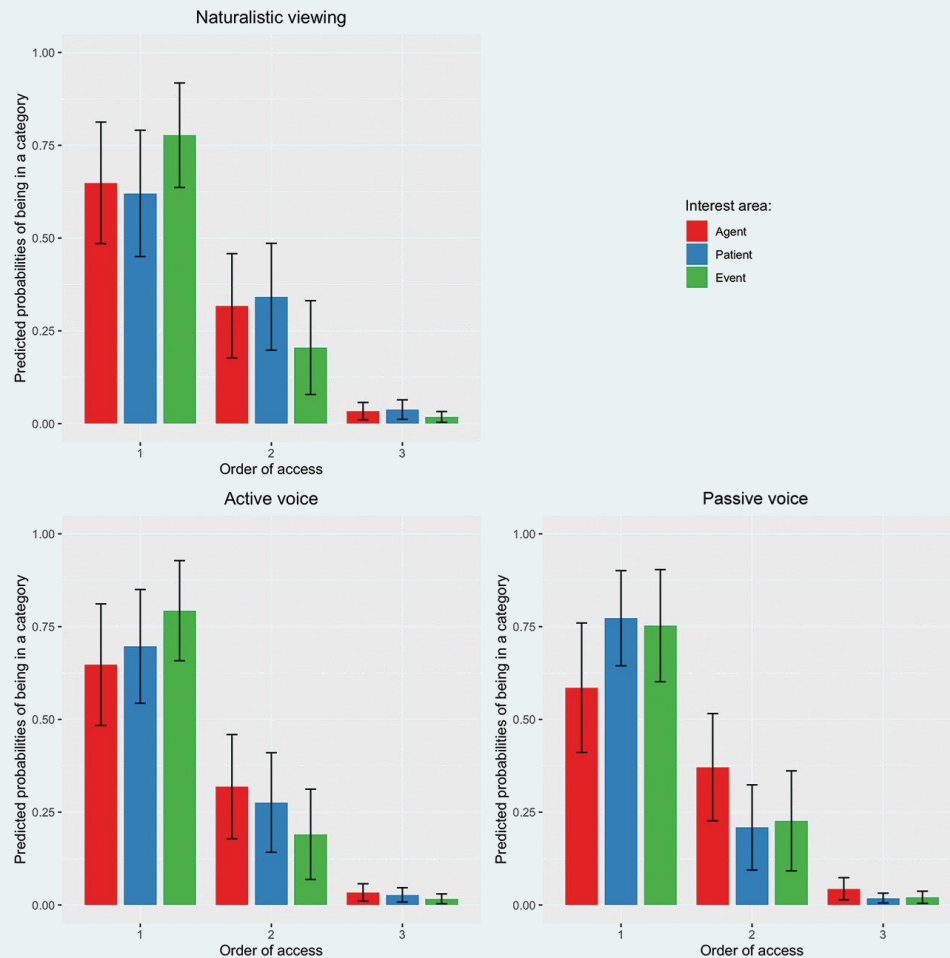
Empirical Methods

- **Eye-tracking** shows that linguistic alternates influence visual perception consistent with construal (Divjak, Milin & Medimorec 2020)
 - Effects of construal compared to naturalistic (image only) viewing
 - Preposition, voice and dative alternations
 - Order and duration of attention



1. (a) The policeman arrested the thief (active)
(b) The thief was arrested by the policeman (passive)

Empirical Methods



Although the Event is significantly more likely to be accessed first in naturalistic viewing, this effect is less pronounced after an active voice description and no longer significant after a passive voice description; after a passive voice description, the Patient is equally likely to be accessed first. (Divjak, Milin & Medimorec 2020: 63)

Applications

- CL goes beyond 'core' areas of linguistic theory to provide frameworks for analysis in applied areas:
 - Cognitive sociolinguistics (Geeraerts, Kristiansen & Peirsman 2010)
 - Cognitive poetics (Stockwell 2020)
 - Cognitive critical discourse analysis (Hart 2024)



Thank you

Selected References

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